

⑨ a $T_1 = \text{Reflection}$ $T_2 = \text{Compression}$

$$T_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \frac{1}{3}(T_2) = \begin{bmatrix} 1/3 & 0 \\ 0 & 1 \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}} \right\} \begin{array}{l} \text{standard} \\ \text{matrices} \\ \text{of} \\ T_1 \text{ \& } T_2 \end{array}$$

$$[T_1 \cdot T_2] = [T_1][T_2] = \begin{bmatrix} 1/3 & 0 \\ 0 & -1 \end{bmatrix}$$

$$[T_2 \cdot T_1] = [T_2][T_1] = \begin{bmatrix} 1/3 & 0 \\ 0 & -1 \end{bmatrix}$$

$\therefore T_1 \cdot T_2 = T_2 \cdot T_1$, the operators commute

b

$T_1 = \text{Reflection}$ $T_2 = \text{Expansion}$

$$T_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad 2(T_2) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} \right\} \begin{array}{l} \text{standard} \\ \text{matrices} \\ \text{of} \\ T_1 \text{ \& } T_2 \end{array}$$

$$[T_1 \cdot T_2] = [T_1][T_2] = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

$$[T_2 \cdot T_1] = [T_2][T_1] = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

$\therefore T_1 \cdot T_2 \neq T_2 \cdot T_1$, the operators don't commute